

Methods for Global Navigation Satellite Systems (GNSS) and Inertial Navigation Systems (INS)

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1 Introduction to Method of Least Squares

This method serves as the basis of many control algorithms. Let assume that we want to estimate a set of scalar variables: $x_1, x_2, \dots, x_n$. However, all we have are the noisy observed (measured) variables $y_1, y_2, \dots, y_s$, where in the general case $s > n$ (basic engineering intuition tells us that we need to have more measurements than unknowns, that is the main reason why $s \geq n$).

Let assume that the observed variables are related to the set of scalar variables that we want to estimate via the following linear model:

$$y_i = \sum_{j=1}^n A_{ij}x_j + \nu_i, \quad i = 1, 2, \dots, s, \quad (1)$$

where $A_{ij} \in \mathbb{R}$ are scalars and $\nu_i \in \mathbb{R}$ is the measurement noise. We emphasize that $\nu_i \in \mathcal{N}(0, \sigma_i^2)$, for $i = 1, 2, \dots, n$ with ν_i and ν_j are statistically independent for $i \neq j$.

Equations (1) can be compactly written as follows:

$$\mathbf{y} = A\mathbf{x} + \boldsymbol{\nu}, \quad (2)$$

where $\mathbf{y}, \boldsymbol{\nu} \in \mathbb{R}^s$ and $A \in \mathbb{R}^{s \times n}$.

In order to formulate the least-squares problem, we need to quantify how accurate is our estimate of scalar variables. We introduce the measurement error:

$$e_i = y_i - \sum_{j=1}^n A_{ij}x_j, \quad i = 1, 2, \dots, s. \quad (3)$$

The compact form is given by

$$\mathbf{e} = \mathbf{y} - A\mathbf{x}. \quad (4)$$

The main idea of the least-squares method is to penalize the weighted sum of squares of the errors. This weighted sum is called a cost function. In the case of the least-squares method, the cost function has the following form:

$$J = \sum_{i=1}^s \frac{e_i^2}{\sigma_i^2}. \quad (5)$$

The idea by divided by the noise variance is deeply rooted in engineering intuition and common sense. Namely, if a noise ν_i is significant, then its variance will be large. Consequently, we are less confident that the measurement y_i will contain useful information. in other words, we want to put less emphasis on the squared measurement error e_i^2 in the cost function. We can do that by dividing the square of measurement error by variance. If the variance has a large value, then the term $\frac{e_i^2}{\sigma_i^2}$ will be relatively small. On the other hand, if the noise is not significant, then the term $\frac{e_i^2}{\sigma_i^2}$ will accordingly increase its influence in the overall sum that is represented by J .

The cost function (5) can be written as:

$$J = \mathbf{e}^\top W \mathbf{e} = (\mathbf{y} - A\mathbf{x})^\top W (\mathbf{y} - A\mathbf{x}), \quad (6)$$

where W is a diagonal weighted matrix defined as follows:

$$W = \begin{pmatrix} \frac{1}{\sigma_1^2} & 0 & 0 \cdots & 0 \\ 0 & \frac{1}{\sigma_2^2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\sigma_s^2} \end{pmatrix}. \quad (7)$$

The least-squares problem can be formally represented as follows:

$$\min_x \mathbf{e}^\top W \mathbf{e}. \quad (8)$$

In order to find the minimum of this cost function, we need to compute its gradient and set it to zero. By solving this equation which is often referred to as the normal equation, we can find an optimal value of $\mathbf{x}$. We have first:

$$J = \mathbf{y}^\top W \mathbf{y} - \mathbf{y}^\top W A \mathbf{x} - \mathbf{x}^\top A^\top W \mathbf{y} + \mathbf{x}^\top A^\top W A \mathbf{x} \quad (9)$$

Then, we can express the gradient of J as follows:

$$\nabla J = -2A^\top W \mathbf{y} + 2A^\top W A \mathbf{x}. \quad (10)$$

By setting the gradient to zero, we obtain the system of normal equations:

$$-A^\top W \mathbf{y} + A^\top W A \mathbf{x} = 0. \quad (11)$$

Under the assumption that the matrix $A^\top W A$ is invertible, we obtain the solution of the least-squares problem.

$$\hat{\mathbf{x}} = (A^\top W A)^{-1} A^\top W \mathbf{y}, \quad (12)$$

where $(A^\top W A)^{-1}$ is the inverse of $A^\top W A$, and $\hat{\mathbf{x}}$ is used to denote the solution of the least-squares problem.

2 Recursive Least Squares Method

The motivation comes from the fact that these equations are used in the derivation of the **Kalman Filter equations**. The **recursive least squares** method is the basis of many other algorithms, such as **adaptive control algorithms**. Consequently, it is of paramount importance to properly understand the recursive least-squares method. **The derivation of the Kalman filter equations on the basis of the recursive least-square equations is arguably simpler and easier to understand than the derivation based on the other methods or approaches.**

We recall that, usually, in the least squares estimation problem defined by :

$$\mathbf{y} = A \mathbf{x} + \boldsymbol{\nu},$$

$n \ll s$, that is the number of measurements s is usually much larger than the number of variables to be estimated. **Under the assumption that the matrix A has full column rank and $s \geq n$** , the least squares solution is found by solving the following optimization problem

$$\min_x \|\mathbf{y} - A \mathbf{x}\|_2^2. \quad (13)$$

There are a number of computationally efficient approaches for solving (13). In the previous section, for brevity and simplicity, we use the solution that is not the most efficient one to be computed. This solution is given by the following equation

$$\hat{\mathbf{x}} = (A^\top A)^{-1} A^\top \mathbf{y}. \quad (14)$$

There are at least two issues with the formulation and solution of the least squares problem:

1. The least-squares problem formulation and its solutions assume that all measurements are available at a certain time instant. **However, this often might not be the case in the practice.** Also, we would like to update the solution as new measurements arrive in order to have an adaptive approach for updating the solution.
2. When the number of measurements s is extremely large, the solutions of the least squares problem are difficult to compute. That is, the computational and memory complexities of the solution are very high.

This facts motivate the development of the recursive least squares that is presented in the sequel.

2.1 Recursive Least Squares Method: Complete Derivation From Scratch

Let us assume that at the discrete-time instant k , the set of measurements grouped in the vector $\mathbf{y}_k \in \mathbb{R}^l$ becomes available. Let us assume that these measurements are related to the vector $\mathbf{x}$ that we want to estimate by the following equation

$$\mathbf{y}_k = A_k \mathbf{x} + \boldsymbol{\nu}_k, \quad (15)$$

where $A_k \in \mathbb{R}^{l \times n}$, and $\boldsymbol{\nu}_k \in \mathbb{R}^l$ is the measurement noise vector. We assume that the covariance of the measurement noise is given by

$$\mathbb{E}[\boldsymbol{\nu}_k^\top \boldsymbol{\nu}_k] = R_k, \quad (16)$$

and we assume that the mean of $\boldsymbol{\nu}_k$, $\mathbb{E}[\boldsymbol{\nu}_k] = \mathbf{0}$.

The recursive least-squares method that we derive in this section, has the following form:

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{k-1} + K_k(\mathbf{y}_k - A_k \hat{\mathbf{x}}_{k-1}), \quad (17)$$

where $\hat{\mathbf{x}}_k$ and $\hat{\mathbf{x}}_{k-1}$ are the estimates of the vector $\mathbf{x}$ at the discrete-time instants k and $k - 1$, and $K_k \in \mathbb{R}^{n \times l}$ is the gain matrix that we need to determine.

The equation (17) updates the estimate of $\mathbf{x}$ at the time instant k on the basis of the estimate $\hat{\mathbf{x}}_{k-1}$ at the previous time instant $k - 1$ and on the basis of the measurement $\mathbf{y}_k$ obtained at the time instant k , as well as on the basis of the gain matrix K_k computed at the time instant k .

Our goal is to derive the formulas for computing the gain matrix K_k .

We assume that the vector $\hat{\mathbf{x}}_k$ is partitioned as followed:

$$\hat{\mathbf{x}}_k = \begin{pmatrix} \hat{x}_{1,k} \\ \hat{x}_{2,k} \\ \vdots \\ \hat{x}_{n,k} \end{pmatrix}.$$

This partitioning directly corresponds to the partitioning of the vector $\mathbf{x}$:

$$\hat{\mathbf{x}} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

Next, we introduce **the estimation error vector** $\boldsymbol{\varepsilon}_k$ and the following partitioning:

$$\boldsymbol{\varepsilon}_k = \begin{pmatrix} \varepsilon_{1,k} \\ \varepsilon_{2,k} \\ \vdots \\ \varepsilon_{n,k} \end{pmatrix} = \mathbf{x} - \hat{\mathbf{x}}_k = \begin{pmatrix} x_1 - \hat{x}_{1,k} \\ x_2 - \hat{x}_{2,k} \\ \vdots \\ x_n - \hat{x}_{n,k} \end{pmatrix}.$$

The gain K_k is computed by minimizing the sum of variances of the estimation errors:

$$W_k = \mathbb{E}[\varepsilon_{1,k}^2] + \mathbb{E}[\varepsilon_{2,k}^2] + \cdots + \mathbb{E}[\varepsilon_{n,k}^2] \quad (18)$$

Next, we show that this cost function, can be represented as follows:

$$W_k = \text{Tr}[P_k], \quad (19)$$

where $\text{Tr}[\cdot]$ is the notation for the matrix trace, and P_k **is the estimation error covariance matrix that is defined by**

$$P_k = \mathbb{E}[\boldsymbol{\varepsilon}_k \boldsymbol{\varepsilon}_k^\top]. \quad (20)$$

That is, **the gain matrix K_k is found by minimizing the trace of the estimation error covariance matrix.**

To show this, let us first expand the following expression:

$$\boldsymbol{\varepsilon}_k \boldsymbol{\varepsilon}_k^\top = \begin{pmatrix} \varepsilon_{1,k} \\ \varepsilon_{2,k} \\ \vdots \\ \varepsilon_{n,k} \end{pmatrix} (\varepsilon_{1,k} \quad \varepsilon_{2,k} \quad \cdots \quad \varepsilon_{n,k}) = \begin{pmatrix} \varepsilon_{1,k}^2 & \varepsilon_{1,k}\varepsilon_{2,k} & \varepsilon_{1,k}\varepsilon_{3,k} & \cdots & \varepsilon_{1,k}\varepsilon_{n,k} \\ \varepsilon_{1,k}\varepsilon_{2,k} & \varepsilon_{2,k}^2 & \varepsilon_{2,k}\varepsilon_{3,k} & \cdots & \varepsilon_{2,k}\varepsilon_{n,k} \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ \varepsilon_{1,k}\varepsilon_{n,k} & \varepsilon_{2,k}\varepsilon_{n,k} & \varepsilon_{3,k}\varepsilon_{n,k} & \cdots & \varepsilon_{n,k}^2 \end{pmatrix}.$$

From the last expression, we have

$$P_k = \mathbb{E}[\boldsymbol{\varepsilon}_k \boldsymbol{\varepsilon}_k^\top] = \begin{pmatrix} \mathbb{E}[\varepsilon_{1,k}^2] & \mathbb{E}[\varepsilon_{1,k}\varepsilon_{2,k}] & \mathbb{E}[\varepsilon_{1,k}\varepsilon_{3,k}] & \cdots & \mathbb{E}[\varepsilon_{1,k}\varepsilon_{n,k}] \\ \mathbb{E}[\varepsilon_{1,k}\varepsilon_{2,k}] & \mathbb{E}[\varepsilon_{2,k}^2] & \mathbb{E}[\varepsilon_{2,k}\varepsilon_{3,k}] & \cdots & \mathbb{E}[\varepsilon_{2,k}\varepsilon_{n,k}] \\ \vdots & \vdots & \vdots & \cdots & \vdots \\ \mathbb{E}[\varepsilon_{1,k}\varepsilon_{n,k}] & \mathbb{E}[\varepsilon_{2,k}\varepsilon_{n,k}] & \mathbb{E}[\varepsilon_{3,k}\varepsilon_{n,k}] & \cdots & \mathbb{E}[\varepsilon_{n,k}^2] \end{pmatrix}.$$

Next, by taking the matrix trace of the last expression, we obtain:

$$\text{Tr}[P_k] = \mathbb{E}[\varepsilon_{1,k}^2] + \mathbb{E}[\varepsilon_{2,k}^2] + \cdots + \mathbb{E}[\varepsilon_{n,k}^2] = W_k.$$

Using the expression of $\hat{\mathbf{x}}_k$ from (17), we have:

$$\begin{aligned} \boldsymbol{\varepsilon}_k &= \mathbf{x} - \hat{\mathbf{x}}_k \\ &= \mathbf{x} - \hat{\mathbf{x}}_{k-1} - K_k(\mathbf{y}_k - A_k \hat{\mathbf{x}}_{k-1}) \\ &= \mathbf{x} - \hat{\mathbf{x}}_{k-1} - K_k(A_k \mathbf{x} + \boldsymbol{\nu}_k - A_k \hat{\mathbf{x}}_{k-1}) \\ &= (\mathbb{I} - K_k A_k)(\mathbf{x} - \hat{\mathbf{x}}_{k-1}) - K_k \boldsymbol{\nu}_k \\ &= (\mathbb{I} - K_k A_k) \boldsymbol{\varepsilon}_{k-1} - K_k \boldsymbol{\nu}_k, \end{aligned}$$

Next, we have

$$\begin{aligned}
\boldsymbol{\varepsilon}_k \boldsymbol{\varepsilon}_k^\top &= ((\mathbb{1} - K_k A_k) \boldsymbol{\varepsilon}_{k-1} - K_k \boldsymbol{\nu}_k) ((\mathbb{1} - K_k A_k) \boldsymbol{\varepsilon}_{k-1} - K_k \boldsymbol{\nu}_k)^\top \\
&= ((\mathbb{1} - K_k A_k) \boldsymbol{\varepsilon}_{k-1} - K_k \boldsymbol{\nu}_k) (\boldsymbol{\varepsilon}_{k-1}^\top (\mathbb{1} - K_k A_k)^\top - \boldsymbol{\nu}_k^\top K_k^\top) \\
&= (\mathbb{1} - K_k A_k) \boldsymbol{\varepsilon}_{k-1} \boldsymbol{\varepsilon}_{k-1}^\top (\mathbb{1} - K_k A_k)^\top - (\mathbb{1} - K_k A_k) \boldsymbol{\varepsilon}_{k-1}^\top \boldsymbol{\nu}_k^\top K_k^\top \\
&\quad - K_k \boldsymbol{\nu}_k \boldsymbol{\varepsilon}_{k-1}^\top (\mathbb{1} - K_k A_k)^\top + K_k \boldsymbol{\nu}_k \boldsymbol{\nu}_k^\top K_k^\top
\end{aligned}$$

Applying the expectation operator to $\boldsymbol{\varepsilon}_k \boldsymbol{\varepsilon}_k^\top$, we obtain:

$$\begin{aligned}
P_k &= \mathbb{E}[\boldsymbol{\varepsilon}_k \boldsymbol{\varepsilon}_k^\top] \\
&= (\mathbb{1} - K_k A_k) \mathbb{E}[\boldsymbol{\varepsilon}_{k-1} \boldsymbol{\varepsilon}_{k-1}^\top] (\mathbb{1} - K_k A_k)^\top - (\mathbb{1} - K_k A_k) \mathbb{E}[\boldsymbol{\varepsilon}_{k-1}^\top \boldsymbol{\nu}_k^\top] K_k^\top \\
&\quad - K_k \mathbb{E}[\boldsymbol{\nu}_k \boldsymbol{\varepsilon}_{k-1}^\top] (\mathbb{1} - K_k A_k)^\top + K_k \mathbb{E}[\boldsymbol{\nu}_k \boldsymbol{\nu}_k^\top] K_k^\top.
\end{aligned}$$

We have

$$P_{k-1} = \mathbb{E}[\boldsymbol{\varepsilon}_{k-1} \boldsymbol{\varepsilon}_{k-1}^\top].$$

Then, since $\boldsymbol{\nu}_k$ and $\boldsymbol{\varepsilon}_{k-1}$ are independent (measurement noise at the time instant k is independent from the estimation error at the time instant $k-1$), and due to the fact that $\boldsymbol{\nu}_k$ is zero-mean:

$$\mathbb{E}[\boldsymbol{\varepsilon}_{k-1} \boldsymbol{\nu}_k^\top] = \mathbb{E}[\boldsymbol{\varepsilon}_{k-1}] \mathbb{E}[\boldsymbol{\nu}_k^\top] = 0; \quad \mathbb{E}[\boldsymbol{\nu}_k \boldsymbol{\varepsilon}_{k-1}^\top] = \mathbb{E}[\boldsymbol{\varepsilon}_{k-1}^\top] \mathbb{E}[\boldsymbol{\nu}_k] = 0; \quad \mathbb{E}[\boldsymbol{\nu}_k \boldsymbol{\nu}_k^\top] = R_k.$$

We obtain:

$$P_k = (\mathbb{1} - K_k A_k) P_{k-1} (\mathbb{1} - K_k A_k)^\top + K_k R_k K_k^\top. \quad (21)$$

Let us expand the expression of P_k . We get:

$$P_k = P_{k-1} - P_{k-1} A_k^\top K_k^\top - K_k A_k P_{k-1} + K_k A_k P_{k-1} A_k^\top K_k^\top + K_k R_k K_k^\top$$

$$W_k = \text{Tr}[P_k] = \text{Tr}[P_{k-1}] - \text{Tr}[P_{k-1} A_k^\top K_k^\top] - \text{Tr}[K_k A_k P_{k-1}] + \text{Tr}[K_k A_k P_{k-1} A_k^\top K_k^\top] + \text{Tr}[K_k R_k K_k^\top].$$

We find the gain matrix K_k by optimizing the cost function W_k . That is, we find the value of K_k by solving the following equation:

$$\frac{\partial W_k}{\partial K_k} = 0. \quad (22)$$

To compute the derivatives of matrix traces, we use the following formulas:

$$\frac{\partial \text{Tr}[A X^\top B]}{\partial X} = B A; \quad \frac{\partial \text{Tr}[A X B]}{\partial X} = A^\top B^\top; \quad \frac{\partial \text{Tr}[X B X^\top B]}{\partial X} = X B^\top + X B. \quad (23)$$

Using these formulas, we obtain:

$$\begin{aligned}
\frac{\partial \text{Tr}[P_{k-1}]}{\partial K_k} &= 0; \quad \frac{\partial \text{Tr}[P_{k-1} A_k^\top K_k^\top]}{\partial K_k} = P_{k-1} A_k^\top; \quad \frac{\partial \text{Tr}[K_k A_k P_{k-1}]}{\partial K_k} = P_{k-1} A_k^\top; \\
\frac{\partial \text{Tr}[K_k A_k P_{k-1} A_k^\top K_k^\top]}{\partial K_k} &= 2 K_k A_k P_{k-1} A_k^\top; \quad \frac{\partial \text{Tr}[K_k R_k K_k^\top]}{\partial K_k} = 2 K_k R_k.
\end{aligned}$$

Plug all this results in (22), gives:

$$-2P_{k-1}A_k^\top + 2K_k A_k P_{k-1} A_k^\top + 2K_k R_k = 0.$$

From the last equation, we have

$$K_k(R_k + A_k P_{k-1} A_k^\top) = P_{k-1} A_k^\top.$$

By solving the last equation we obtain the final expression for the gain matrix K_k

$$K_k = P_{k-1} A_k^\top (R_k + A_k P_{k-1} A_k^\top)^{-1}. \quad (24)$$

Before we summarize the recursive least-squares method, we derive an alternative form of the estimation error covariance matrix propagation equation. First, we can write the gain matrix, as follows:

$$K_k = P_{k-1} A_k^\top L_k^{-1}; \quad L_k = R_k + A_k P_{k-1} A_k^\top,$$

Here, we should observe that the matrix L_k and its inverse L_k^{-1} are symmetric, that is:

$$L_k = L_k^\top, \quad (L_k^{-1})^\top = L_k^{-1}.$$

Substitute (24) in P_k , we obtain:

$$\begin{aligned} P_k &= P_{k-1} - P_{k-1} A_k^\top L_k^{-1} A_k P_{k-1} - P_{k-1} A_k^\top L_k^{-1} A_k P_{k-1} \\ &\quad + P_{k-1} A_k^\top L_k^{-1} A_k P_{k-1} A_k^\top L_k^{-1} A_k P_{k-1} + P_{k-1} A_k^\top L_k^{-1} R_k L_k^{-1} A_k P_{k-1} \end{aligned}$$

By adding the second and the third terms and by grouping the last two terms, we obtain:

$$P_k = P_{k-1} - 2P_{k-1} A_k^\top L_k^{-1} A_k P_{k-1} + P_{k-1} A_k^\top L_k^{-1} \underbrace{(A_k P_{k-1} A_k^\top + R_k)}_{L_k} L_k^{-1} A_k P_{k-1} \quad (25)$$

$$= P_{k-1} - 2P_{k-1} A_k^\top L_k^{-1} A_k P_{k-1} + P_{k-1} A_k^\top L_k^{-1} A_k P_{k-1} \quad (26)$$

$$= P_{k-1} - \underbrace{P_{k-1} A_k^\top L_k^{-1} A_k P_{k-1}}_{K_k} \quad (27)$$

$$= P_{k-1} - K_k A_k P_{k-1} \quad (28)$$

$$P_k = (\mathbf{1} - K_k A_k) P_{k-1}. \quad (29)$$

That is, an alternative expression for the propagation of the estimation error covariance matrix given by

$$P_k = (\mathbf{1} - K_k A_k) P_{k-1} \quad (30)$$

Consequently, the recursive least squares method consists of the following three equations

1. Gain matrix update:

$$K_k = P_{k-1} A_k^\top (R_k + A_k P_{k-1} A_k^\top)^{-1} \quad (31)$$

2. Estimate update

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{k-1} + K_k (\mathbf{y}_k - A_k \hat{\mathbf{x}}_{k-1}) \quad (32)$$

3. Propagation of the estimation error covariance matrix by using this equation

$$P_k = (\mathbb{1} - K_k A_k) P_{k-1} (\mathbb{1} - K_k A_k)^\top + K_k R_k K_k^\top \quad (33)$$

or this equation

$$P_k = (\mathbb{1} - K_k A_k) P_{k-1} \quad (34)$$

If no apriori knowledge of the estimate is known, then this method is initialized with the estimation error covariance matrix that is equal to $P_0 = c\mathbb{1}$, where c is large number, and with a random value of $\mathbf{x}_0$.

3 Time propagation of state vector expectation and state covariance matrix of linear dynamic systems - Intro to Kalman Filtering

In this section, we explain how the expected value of the state vector and state covariance matrix of linear dynamic systems are propagated in discrete-time. These equations, together with the equations describing the recursive least-squares method are important for deriving the Kalman filter equations.

3.1 Linear dynamical system affected by the process noise (disturbances)

We are considering the following state equation:

$$\mathbf{x}_k = A_{k-1} \mathbf{x}_{k-1} + B_{k-1} \mathbf{u}_{k-1} + \mathbf{w}_{k-1}, \quad (35)$$

where:

- $\mathbf{x}_k \in \mathbb{R}^n$ is the state vector
- $\mathbf{u}_{k-1} \in \mathbb{R}^m$ is the control input vector
- $\mathbf{w}_{k-1} \in \mathbb{R}^n$ is the disturbance or process noise vector. We assume that $\mathbf{w}_{k-1}$ is Gaussian, zero-mean, white noise with the covariance matrix given by $Q_k = \mathbb{E}[\mathbf{w}_k \mathbf{w}_k^\top]$, where $\mathbb{E}[\cdot]$ is the mathematical expectation operator.
- $A_{k-1} \in \mathbb{R}^{n \times n}$ and $B_{k-1} \in \mathbb{R}^{n \times m}$ are state and input matrices.

Remark 3.1. *If the system is time-invariant, then the matrices A_k , B_k , and Q_k are constant, that is $A_k = A$, $B_k = B$, and $Q_k = Q$. The process disturbance $\mathbf{w}_k$ can take into account unmodeled dynamics that is simply assumed as a stochastic disturbance, and/or stochastic disturbances affecting the state vector.*

3.2 State vector expectation propagation

Let us apply the expectation operator to the equation (35), as the result, we obtain:

$$\mathbb{E}[\mathbf{x}_k] = \mathbb{E}[A_{k-1} \mathbf{x}_{k-1}] + \mathbb{E}[B_{k-1} \mathbf{u}_{k-1}] + \mathbb{E}[\mathbf{w}_{k-1}].$$

For notation brevity, we introduce the following notation: $\mathbb{E}[\mathbf{x}_k] = \bar{\mathbf{x}}_k$.

Taking into account that $\mathbb{E}[\mathbf{w}_{k-1}] = 0$ (the disturbance vector os zero mean), **we obtain our final equation for state vector expectation propagation over discrete-time:**

$$\bar{\mathbf{x}}_k = A\bar{\mathbf{x}}_{k-1} + B_{k-1}\mathbf{u}_{k-1} \quad (36)$$

3.3 State covariance matrix propagation

We define the state covariance matrix as follows:

$$P_k = \mathbb{E}[(\mathbf{x}_k - \bar{\mathbf{x}}_k)(\mathbf{x}_k - \bar{\mathbf{x}}_k)^\top]. \quad (37)$$

Consequently, we first need to expend the expression: $(\mathbf{x}_k - \bar{\mathbf{x}}_k)(\mathbf{x}_k - \bar{\mathbf{x}}_k)^\top$ and compute the expectation.

$$\begin{aligned} (\mathbf{x}_k - \bar{\mathbf{x}}_k)(\mathbf{x}_k - \bar{\mathbf{x}}_k)^\top &= (A_{k-1}(\mathbf{x}_{k-1} - \bar{\mathbf{x}}_{k-1}) + \mathbf{w}_{k-1})(A_{k-1}(\mathbf{x}_{k-1} - \bar{\mathbf{x}}_{k-1}) + \mathbf{w}_{k-1})^\top \\ &= A_{k-1}(\mathbf{x}_{k-1} - \bar{\mathbf{x}}_{k-1})(\mathbf{x}_{k-1} - \bar{\mathbf{x}}_{k-1})^\top A_{k-1}^\top + A_{k-1}(\mathbf{x}_{k-1} - \bar{\mathbf{x}}_{k-1})\mathbf{w}_{k-1}^\top \\ &\quad + \mathbf{w}_{k-1}(\mathbf{x}_{k-1} - \bar{\mathbf{x}}_{k-1})^\top A_{k-1}^\top + \mathbf{w}_{k-1}\mathbf{w}_{k-1}^\top \end{aligned}$$

Since $(\mathbf{x}_k - \bar{\mathbf{x}}_k)$ is uncorrelated with $\mathbf{w}_{k-1}$, we have:

$$\mathbb{E}[A_{k-1}(\mathbf{x}_{k-1} - \bar{\mathbf{x}}_{k-1})\mathbf{w}_{k-1}^\top] = 0, \quad \mathbb{E}[\mathbf{w}_{k-1}(\mathbf{x}_{k-1} - \bar{\mathbf{x}}_{k-1})^\top A_{k-1}^\top] = 0.$$

We also have:

$$\mathbb{E}[\mathbf{w}_{k-1}\mathbf{w}_{k-1}^\top] = Q_{k-1}, \quad P_{k-1} = \mathbb{E}[(\mathbf{x}_{k-1} - \bar{\mathbf{x}}_{k-1})(\mathbf{x}_{k-1} - \bar{\mathbf{x}}_{k-1})^\top].$$

We obtain the final equation for the propagation of the state covariance matrix:

$$P_k = A_{k-1}P_{k-1}A_{k-1}^\top + Q_{k-1} \quad (38)$$

4 Kalman filter - derivation of the Kalman filter by using recursive least-squares method

In this section, we present a simple derivation of the Kalman filter equations.

4.1 Model and Model Assumptions

We are considering the following state-space model of a dynamic system:

$$\mathbf{x}_k = A_{k-1}\mathbf{x}_{k-1} + B_{k-1}\mathbf{u}_{k-1} + \mathbf{w}_{k-1}, \quad \mathbf{y}_{k-1} = C_k\mathbf{x}_k + \nu_k \quad (39)$$

where:

- k is the discrete-time instant
- $\mathbf{x}_k \in \mathbb{R}^n$ is the state vector at the discrete time instant k
- $\mathbf{u}_{k-1} \in \mathbb{R}^m$ is the control input vector
- $\mathbf{w}_{k-1} \in \mathbb{R}^n$ is the disturbance or process noise vector. We assume that $\mathbf{w}_k$ is white, zero mean, uncorrelated, with the covariance matrix $Q_k = \mathbb{E}[\mathbf{w}_k\mathbf{w}_k^\top]$.

- $A_{k-1} \in \mathbb{R}^{n \times n}$ and $B_{k-1} \in \mathbb{R}^{n \times m}$ are state and input matrices.
- $C_k \in \mathbb{R}^{r \times n}$ is the output matrix
- $\mathbf{y}_k \in \mathbb{R}^r$ is the output vector (observed measurements)
- $\mathbf{v}_k \in \mathbb{R}^r$ is the measurement noise vector. We assume that $\mathbf{v}_k$ is white, zero mean, uncorrelated, with the covariance matrix given by $R_k = \mathbb{E}[\mathbf{v}_k \mathbf{v}_k^\top]$

This is a time-varying system. However, if the system is time-invariant, then all the system matrices are constant.

We assume that the covariance matrices R_k, Q_k , as well as systems matrices A_k, B_k , and C_k are known.

The goal is to design an observer (estimator) that will estimate the state vector $\mathbf{x}_k$.

4.2 A priori and a posteriori state estimates, errors, and covariance matrices

In Kalman filtering, we have two important estimates: **a priori state estimates** and **a posteriori state estimates**.

The **a priori state estimate** of the state vector $\mathbf{x}_k$ is obtained implicitly on the basis of the output vector measurements $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_{k-1}$ (as well as on some other information that is not relevant for this discussion). That is, **the a priori state estimate is obtained on the basis of the past measurements up to the current discrete-time instant k and NOT on the basis of the measurements $\mathbf{y}_k$ at the current discrete-time instant k** . The a priori state estimate is denoted by: $\hat{\mathbf{x}}_k^-$ (a priori state estimate), where *the hat notation* denotes an estimate, and where the minus superscript denotes the a priori state estimate. **The minus superscript originates from the fact that this estimate is obtained before we process the measurement $\mathbf{y}_k$ at the time instant k .**

The a posteriori estimate of the state $\mathbf{x}_k$ is obtained implicitly on the basis of the measurements $\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_{k-1}, \mathbf{y}_k$. The a posteriori estimate is denoted by: $\hat{\mathbf{x}}_k^+$ (a posteriori state estimate), **where the plus superscript in the state estimate notation denotes the fact that the a posteriori estimate is obtained by processing the measurement $\mathbf{y}_k$ obtained at the estimate time instant k** . Here, we also need to introduce covariance matrices of the a priori and a posteriori estimation errors.

The covariance matrices of the a priori and a posteriori estimation errors are defined as follows:

$$P_k^- = \mathbb{E}[(\mathbf{x}_k - \hat{\mathbf{x}}_k^-)(\mathbf{x}_k - \hat{\mathbf{x}}_k^-)^\top] \quad (40)$$

$$P_k^+ = \mathbb{E}[(\mathbf{x}_k - \hat{\mathbf{x}}_k^+)(\mathbf{x}_k - \hat{\mathbf{x}}_k^+)^\top] \quad (41)$$

where P_k^- is the covariance matrix of the a priori estimation error and P_k^+ is the covariance matrix of the a posteriori estimation error.

4.3 Derivation of Kalman equations from the recursive-least squares method

The following figure illustres how the Kalman filter works in pratice:

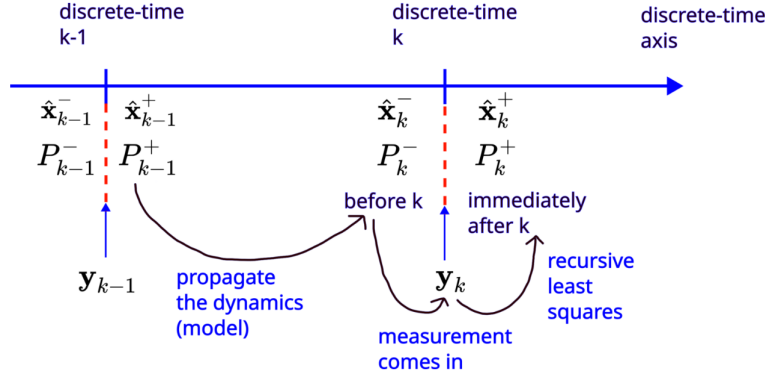


Figure 1: Illustration of Kalman filtering equations

Immediately after the discrete-time $k - 1$, we compute the a posteriori state $\mathbf{x}_{k-1}^+$ and the covariance matrix of the estimation error P_{k-1}^+ . Then, we propagate these quantities through the equations describing the system dynamics and covariance matrix propagation (that is also derived on the basis of the system dynamics). That is, we propagate the a posteriori state and covariance through our model. By propagating these quantities, we obtain the a priori state estimate $\mathbf{x}_k^-$ and the covariance matrix of the estimation error P_k^- for the step k . Then, after the measurement vector $\mathbf{y}_k$ is observed at the discrete-time step k , we use this measurement and the recursive-least squares method to compute the a posteriori estimate $\mathbf{x}_k^+$ and the covariance matrix of the estimation error P_k^+ for the time k .

Now that we have a general idea of how Kalman filtering works, let us now derive the Kalman filter equations. At the initial time instant $k = 0$, we need to choose an initial guess of the state estimate. Let this chosen initial guess of the state estimate be denoted by $\hat{\mathbf{x}}_0$. This will be our initial a posteriori state estimate, that is: $\hat{\mathbf{x}}_0^+ = \hat{\mathbf{x}}_0$.

Then, the question is how to compute the a priori estimate $\hat{\mathbf{x}}_1$ for the time instant 1, before the measurements are taken?

The natural answer is that the system states, as well as the estimates, need to satisfy the system dynamics (39), and consequently,

$$\hat{\mathbf{x}}_1 = A_0 \hat{\mathbf{x}}_0^+ + B_0 \mathbf{u}_0, \quad (42)$$

where we excluded the disturbance part since the disturbance vector is not known. Besides the initial guess of the estimate, we also need to select an initial guess of the covariance matrix of the estimation error. That is, we need P_0^+ . If we have perfect knowledge about the initial state, then we select P_0^+ as a zero matrix, that is $P_0^+ = 0 * \mathbf{1}$, where $\mathbf{1}$ is an identity matrix.

This is because the covariance matrix of the estimation error is the measure of uncertainty, and if we have perfect knowledge, then the measure of uncertainty should be zero. On the other hand, if we do not have any a priori knowledge of the initial state, then

$P_0^+ = c\mathbb{1}$, where c is a large number. Next, we need to compute the covariance matrix of the a priori state estimation error, that is, we need to compute P_1^- .

We derived the expression for the time propagation of the covariance matrix of the state estimation error:

$$P_k = A_{k-1}P_{k-1}A_{k-1}^\top + Q_{k-1} \quad (43)$$

By using this expression, we obtain the following equation for P_1^-

$$P_1^- = A_0P_0^+A_0^\top + Q_0 \quad (44)$$

So this is the best we can do at the initial time step $k = 0$ before we obtain measurement information. We computed the a priori estimate $\hat{\mathbf{x}}_1^-$ and the covariance matrix P_1^- for the time step $k = 1$, and before the measurement $\mathbf{y}_1$ at the step $k = 1$ has arrived.

At the time instant $k = 1$, the measurement $\mathbf{y}_1$, arrives. Then, the natural question is how to update the a priori estimate $\hat{\mathbf{x}}_1^-$ and the covariance matrix P_1^- ?

To update these quantities, we need to use the recursive least-squares method. We derived the equations of the recursive least-squares method as follows:

1. Update the gain matrix K_k :

$$K_k = P_{k-1}C_k^\top(R_k + C_kP_{k-1}C_k^\top)^{-1}$$

2. Update the estimate:

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{k-1} + K_k(\mathbf{y}_k - C_k\hat{\mathbf{x}}_{k-1})$$

3. Propagate the covariance matrix by using this equation:

$$P_k = (\mathbb{1} - K_kC_k)P_{k-1}(\mathbb{1} - K_kC_k)^\top + K_kR_kK_k^\top$$

or this equation

$$P_k = (\mathbb{1} - K_kC_k)P_{k-1}$$

Now, the equation is how to use these equations to derive the Kalman filter equations?

First, we perform the following substitutions in the recursive least-squares equations (these substitutions will be explained later in the text):

$$P_{k-1} \rightarrow P_k^- \quad (45)$$

$$P_k \rightarrow P_k^+ \quad (46)$$

$$\hat{\mathbf{x}}_{k-1} \rightarrow \hat{\mathbf{x}}_k^- \quad (47)$$

$$\hat{\mathbf{x}}_k \rightarrow \hat{\mathbf{x}}_k^+ \quad (48)$$

Here is the main idea behind these substitutions.

Let us first focus on the equation (45). In the case of recursive least squares method, the covariance before the measurement arrives is P_{k-1} . In the case of the Kalman filter, P_{k-1} becomes P_k^- , since P_k^- is the covariance before the measurement arrived.

Next, let us focus on (46). In the case of the recursive least squares method, the covariance after the measurement arrives is P_k . In the case of the Kalman filter, P_k becomes P_k^+ , since P_k^+ is the updated covariance after the measurement arrived.

Next, let us focus on (47). In the case of the recursive least squares, $\hat{\mathbf{x}}_{k-1}$ is the state estimate before the measurement $\mathbf{y}_k$ arrived. Consequently, in the case of the Kalman filter, $\hat{\mathbf{x}}_{k-1}$ is the state estimate before the measurement $\mathbf{y}_k$ arrived. Consequently, in the case of the Kalman filter, $\hat{\mathbf{x}}_{k-1}$ becomes the a priori estimate $\hat{\mathbf{x}}_k^-$, since $\hat{\mathbf{x}}_k^-$ does not take into account the measurement $\mathbf{y}_k$.

Finally, let us focus on (48). In the case of the recursive least squares method, $\hat{\mathbf{x}}_k$ is the state that takes into account the measurement $\mathbf{y}_k$. Consequently, in the case of the Kalman filter, $\hat{\mathbf{x}}_k$ becomes $\hat{\mathbf{x}}_k^+$, since $\hat{\mathbf{x}}_k^+$ take into account the measurement $\mathbf{y}_k$ obtained at the time instant k .

After these substitutions, we obtain the update equations:

$$K_k = P_k^- C_k^\top (R_k + C_k P_k^- C_k^\top)^{-1} \quad (49)$$

$$\hat{\mathbf{x}}_k^+ = \hat{\mathbf{x}}_k^- + K_k (\mathbf{y}_k - C_k \hat{\mathbf{x}}_k^-) \quad (50)$$

$$P_k^+ = (\mathbb{1} - K_k C_k) P_k^- (\mathbb{1} - K_k C_k)^\top + K_k R_k K_k^\top = (\mathbb{1} - K_k C_k) P_k^- \quad (51)$$

Now, let us go back at the time instant $k = 1$. By using the newly established update equations, we update $\mathbf{x}_1^-$ and P_1^- , as follows

$$K_1 = P_1^- C_1^\top (R_1 + C_1 P_1^- C_1^\top)^{-1} \quad (52)$$

$$\hat{\mathbf{x}}_1^+ = \hat{\mathbf{x}}_1^- + K_1 (\mathbf{y}_1 - C_1 \hat{\mathbf{x}}_1^-) \quad (53)$$

$$P_1^+ = (\mathbb{1} - K_1 C_1) P_1^- \quad (54)$$

As the result, we obtained P_1^+ and the a posteriori estimate $\hat{\mathbf{x}}_1^+$.

Next, similarly to initial step, we propagate $\hat{\mathbf{x}}_1^-$ through the system dynamics (model of the system), and P_1^+ through covariance propagation equation:

$$\hat{\mathbf{x}}_2^- = A_1 \hat{\mathbf{x}}_1^+ + B_1 \mathbf{u}_1 \quad (55)$$

$$P_2^- = A_1 P_1^+ A_1^\top + Q_1 \quad (56)$$

to obtain $\hat{\mathbf{x}}_2^-$ and P_2^- . After this step, we compute the a posteriori estimate and the covariance matrix matrix of the estimation error by using Eqs.(49), (50), (51) for $k = 2$. **We learned that the two main steps of the Kalman filter are:**

1. Before the measurement arrives, use the system dynamics to propagate the a posteriori estimate and covariance matrix of the estimation error from the step $k-1$, to obtain the a priori estimate and covariance matrix of the estimation error for the step k .
2. At the time step k , use the recursive least squares equations, and the obtained measurement, to compute the a posteriori estimate and the covariance matrix of the estimation error.

4.4 Summary of the derived Kalman filter equations

In the initial step, we select an initial estimate $\hat{\mathbf{x}}_0^+$ and the covariance matrix of the estimation error P_0^+ . Then for $k = 2, 3, \dots$, we perform steps 1 and 2 summarized below.

Step 1: Propagate the a posteriori estimate $\hat{\mathbf{x}}_{k-1}^+$ and the covariance matrix of the estimation error P_{k-1}^+ , through the system dynamics to obtain the a priori state estimate and the covariance matrix of the estimation error for step k :

$$\hat{\mathbf{x}}_k^- = A_{k-1}\hat{\mathbf{x}}_{k-1}^+ + B_{k-1}\mathbf{u}_{k-1} \quad (57)$$

$$P_k^- = A_{k-1}P_{k-1}^+A_{k-1}^\top + Q_{k-1} \quad (58)$$

Step2: Obtain the measurement $\mathbf{y}_k$. Use this measurement, and the priori estimate $\hat{\mathbf{x}}_k^-$ and the covariance matrix of the estimation error P_k^- to compute the a posteriori state and the covariance matrix of the estimation error

$$K_k = P_k^- C_k^\top (R_k + C_k P_k^- C_k^\top)^{-1} \quad (59)$$

$$\hat{\mathbf{x}}_k^+ = \hat{\mathbf{x}}_k^- + K_k(\mathbf{y}_k - C_k \hat{\mathbf{x}}_k^-) \quad (60)$$

$$P_k^+ = (\mathbf{1} - K_k C_k) P_k^- (\mathbf{1} - K_k C_k)^\top + K_k R_k K_k^\top = (\mathbf{1} - K_k C_k) P_k^- \quad (61)$$

Here are a few comments about the derived Kalman filter equations. The matrix K_k is called **the Kalman filter gain matrix**. Then, the equation for the covariance update

$$P_k^+ = (\mathbf{1} - K_k C_k) P_k^- (\mathbf{1} - K_k C_k)^\top + K_k R_k K_k^\top$$

is more numerically stable and robust than the alternative equation

$$P_k^+ = (\mathbf{1} - K_k C_k) P_k^-$$

Consequently, the equation $P_k^+ = (\mathbf{1} - K_k C_k) P_k^- (\mathbf{1} - K_k C_k)^\top + K_k R_k K_k^\top$ is preferable for computing the covariance update. Another important thing to mention is that the Kalman filter gain K_k does not depend on the measurements $\mathbf{y}_k$. This is because the calculations of P_k^- , P_k^+ , and K_k only depend on the system matrices A_{k-1} , C_{k-1} , and Q_k and R_k . This means that the Kalman filter matrices K_k , for $k = 1, 2, 3, \dots$ can be precomputed before the Kalman filter starts operate online. This is important for saving computational time in embedded systems.

5 Extended Kalman Filter

We present the extended Kalman Filter that is used for the state estimation of nonlinear systems.

5.1 General information about extended Kalman Filter

The extended Kalman Filter is a generalization of the linear Kalman Filter for nonlinear dynamical systems in the fairly general mathematical form given by the following state and output equations:

$$\mathbf{x}_k = f_{k-1}(\mathbf{x}_{k-1}, \mathbf{u}_{k-1}) + \mathbf{d}_{k-1} \quad (62)$$

$$\mathbf{y}_k = g_k(\mathbf{x}_k) + \mathbf{q}_k, \quad (63)$$

where

- k is a discrete-time instant. That is, we look at the system behavior at the time instants $0, \Delta t, 2\Delta t, \dots, k\Delta, \dots$, where Δt is the sampling period of the system. That is, before applying the filter, we assume that the signals are sampled in time.
- $\mathbf{x}_k \in \mathbb{R}^n$ is the n -dimensional **state vector** at the discrete-time instant k .
- $\mathbf{u}_k \in \mathbb{R}^m$ is the m -dimensional **input vector** at the discrete-time instant k .
- $\mathbf{d}_k \in \mathbb{R}^n$ is the n -dimensional **disturbance vector** affecting the state and dynamics of the system. This vector is also called the **process noise vector**. We assume that the disturbance vector is white, zero-mean, uncorrelated, with the covariance matrix given by Q_k .
- $\mathbf{y}_k \in \mathbb{R}^r$ is the r -dimensional **output vector**. This vector is also known as the observation vector.
- $\mathbf{q}_k \in \mathbb{R}^r$ is the r -dimensional **measurement noise vector**. We assume that the measurement noise vector is white, zero-mean, uncorrelated, with the covariance matrix given by R_k .
- f_{k-1} is the nonlinear vector function that is allowed to vary from one time instant to another. This function mathematically models the dynamics of the system.
- g_k is the nonlinear vector function that is allowed to vary from one time instant to another. This function mathematically models the relationship between the observed vector and the state of the system.

5.2 Extended Kalman Filter (EKF)

Let us assume that at the discrete-time instant $k-1$ the a posteriori estimate $\hat{\mathbf{x}}_{k-1}^+$ is available. Also, let us assume that the value of $\mathbf{u}_{k-1} = \mathbf{u}_{k-1}^*$ is also known at this time instant. **The first main idea of the extended filter is to linearize the state equation around $\hat{\mathbf{x}}_{k-1}^+$.** So, let us do that. By linearizing the state equation, we obtain:

$$\begin{aligned} \mathbf{x}_k &\approx f_{k-1}(\hat{\mathbf{x}}_{k-1}^+, \mathbf{u}_{k-1}^*) + A_{k-1}(\mathbf{x}_{k-1} - \hat{\mathbf{x}}_{k-1}^+) + \mathbf{d}_{k-1} \\ \mathbf{x}_k &\approx A_{k-1}\mathbf{x}_{k-1} + f_{k-1}(\hat{\mathbf{x}}_{k-1}^+, \mathbf{u}_{k-1}^*) - A_{k-1}\hat{\mathbf{x}}_{k-1}^+ + \mathbf{d}_{k-1} \\ \mathbf{x}_k &\approx A_{k-1}\mathbf{x}_{k-1} + \mathbf{s}_{k-1} + \mathbf{d}_{k-1}, \end{aligned}$$

where

- The matrix A_{k-1} is obtained by linearizing the nonlinear dynamics around $\hat{\mathbf{x}}_{k-1}^+$ and $\mathbf{u}_{k-1}^*$

$$A_{k-1} = \left. \frac{\partial f_{k-1}(\mathbf{x}_{k-1}, \mathbf{u}_{k-1})}{\partial \mathbf{x}_{k-1}} \right|_{\mathbf{x}_{k-1}=\hat{\mathbf{x}}_{k-1}^+, \mathbf{u}_{k-1}=\mathbf{u}_{k-1}^*}, \quad (64)$$

where $\mathbf{u}_{k-1}^*$ is the known value of the input at the time instant $k-1$. This matrix is the **Jacobian** matrix of the function $f_{k-1}(\mathbf{x}_{k-1}, \mathbf{u}_{k-1})$ evaluated at $\hat{\mathbf{x}}_{k-1}^+$ and $\mathbf{u}_{k-1}^*$. The Jacobian matrix is defined by:

$$\frac{\partial f_{k-1}}{\partial \mathbf{x}_{k-1}}(\mathbf{x}_{k-1}, \mathbf{u}_{k-1}) = \begin{pmatrix} \frac{\partial f_{1,k-1}}{\partial x_{1,k-1}} & \frac{\partial f_{1,k-1}}{\partial x_{2,k-1}} & \cdots & \frac{\partial f_{1,k-1}}{\partial x_{n,k-1}} \\ \frac{\partial f_{2,k-1}}{\partial x_{1,k-1}} & \frac{\partial f_{2,k-1}}{\partial x_{2,k-1}} & \cdots & \frac{\partial f_{2,k-1}}{\partial x_{n,k-1}} \\ \vdots & \vdots & \cdots & \vdots \\ \frac{\partial f_{n,k-1}}{\partial x_{1,k-1}} & \frac{\partial f_{n,k-1}}{\partial x_{2,k-1}} & \cdots & \frac{\partial f_{n,k-1}}{\partial x_{n,k-1}} \end{pmatrix}, \quad (65)$$

where

$$f_{k-1}(\mathbf{x}_{k-1}, \mathbf{u}_{k-1}) = \begin{pmatrix} f_{1,k-1} \\ f_{2,k-1} \\ \vdots \\ f_{n,k-1} \end{pmatrix}, \quad \mathbf{x}_{k-1} = \begin{pmatrix} x_{1,k-1} \\ x_{2,k-1} \\ \cdots \\ x_{n,k-1} \end{pmatrix}.$$

- The vector $\mathbf{s}_{k-1}$ contains known quantities at the time instant $k-1$ and is defined as

$$\mathbf{s}_{k-1} = f_{k-1}(\hat{\mathbf{x}}_{k-1}^+, \mathbf{u}_{k-1}^*) - A_{k-1}\hat{\mathbf{x}}_{k-1}^+.$$

The next step is to linearize the output equation. At time $k-1$ we can propagate the state estimate $\hat{\mathbf{x}}_{k-1}^+$ through the nonlinear dynamics to compute $\hat{\mathbf{x}}_k^-$. We do that before the output measurement at the time instant k arrives. **The second main idea of the Kalman filter is to linearize the output equation around the known value for the a priori state estimate $\hat{\mathbf{x}}_k^-$.** Let us do that. First, we have:

$$\mathbf{y}_k = g_k(\mathbf{x}_k) + \mathbf{q}_k.$$

The linearization procedure is given by:

$$\begin{aligned} \mathbf{y}_k &\approx g_k(\hat{\mathbf{x}}_k^-) + C_k(\mathbf{x}_k - \hat{\mathbf{x}}_k^-) + \mathbf{q}_k \\ \mathbf{y}_k &\approx C_k\mathbf{x}_k + g_k(\hat{\mathbf{x}}_k^-) - C_k\hat{\mathbf{x}}_k^- + \mathbf{q}_k \\ \mathbf{y}_k &\approx C_k\mathbf{x}_k + \mathbf{b}_k + \mathbf{q}_k, \end{aligned}$$

where

- The matrix C_k is obtained by linearizing the nonlinear output equation around $\hat{\mathbf{x}}_k^-$

$$C_k = \left. \frac{\partial g_k(\mathbf{x}_k)}{\partial \mathbf{x}_k} \right|_{\mathbf{x}_k=\hat{\mathbf{x}}_k^-}$$

This matrix is the Jacobian matrix of the function $g_k(\mathbf{x}_k)$ evaluated at $\hat{\mathbf{x}}_k^-$. The Jacobian matrix is defined by

$$\frac{\partial g_k}{\partial \mathbf{x}_k}(\mathbf{x}_k) = \begin{pmatrix} \frac{\partial g_{1,k-1}}{\partial x_{1,k}} & \frac{\partial g_{1,k}}{\partial x_{2,k}} & \cdots & \frac{\partial g_{1,k}}{\partial x_{n,k}} \\ \frac{\partial g_{2,k}}{\partial x_{1,k}} & \frac{\partial g_{2,k}}{\partial x_{2,k}} & \cdots & \frac{\partial g_{2,k}}{\partial x_{n,k}} \\ \vdots & \vdots & \cdots & \vdots \\ \frac{\partial g_{n,k}}{\partial x_{1,k}} & \frac{\partial g_{n,k}}{\partial x_{2,k}} & \cdots & \frac{\partial g_{n,k}}{\partial x_{n,k}} \end{pmatrix}, \quad (66)$$

where

$$g_k(\mathbf{x}_k) = \begin{pmatrix} g_{1,k} \\ g_{2,k} \\ \vdots \\ g_{n,k} \end{pmatrix}, \quad \mathbf{x}_k = \begin{pmatrix} x_{1,k} \\ x_{2,k} \\ \cdots \\ x_{n,k} \end{pmatrix}.$$

The vector $\mathbf{b}_k$, is defined as follows:

$$\mathbf{b}_k = g_k(\hat{\mathbf{x}}_k^-) - C_k \hat{\mathbf{x}}_k^-$$

The vector $\mathbf{b}_k$ is completely known.

Let us now summarize the linearized state and output equations:

- Linearized state equation: $\mathbf{x}_k \approx A_{k-1} \mathbf{x}_{k-1} + \mathbf{s}_{k-1} + \mathbf{d}_{k-1}$,
- Linearized output equation: $\mathbf{y}_k \approx C_k \mathbf{x}_k + \mathbf{b}_k + \mathbf{q}_k$

The first main idea of the Extended Kalman Filter is to apply the linear Kalman Filter to these equations. The only divergence from the linear Kalman Filter is to use the nonlinear state equation to propagate the a posteriori estimates in time.

We will use the derived equations and apply them to our case. At the initial discrete-time step $k = 0$ we select **an initial a posteriori estimate** $\hat{\mathbf{x}}_0^+$ and an initial a posteriori covariance matrix of the estimation error P_0^+ . Then for $k = 1, 2, \dots$ we recursively perform the following steps:

- **Step 1 (after the time step $k - 1$ and before the time step k):** At this time step we know the a posteriori estimate $\hat{\mathbf{x}}_{k-1}^+$ and the a posteriori covariance matrix P_{k-1}^+ from the time step $k - 1$. First, we compute the linearized matrix A_{k-1} :

$$A_{k-1} = \left. \frac{\partial f_{k-1}(\mathbf{x}_{k-1}, \mathbf{u}_{k-1})}{\partial \mathbf{x}_{k-1}} \right|_{\mathbf{x}_{k-1}=\hat{\mathbf{x}}_{k-1}^+, \mathbf{u}_{k-1}=\mathbf{u}_{k-1}^*}$$

Then, we compute the a priori estimate $\hat{\mathbf{x}}_k^-$ and the a priori covariance matrix P_k^- for the upcoming time step k by propagating $\hat{\mathbf{x}}_{k-1}^+$ and P_{k-1}^+ through the nonlinear dynamics and the covariance equation

$$\begin{aligned} \hat{\mathbf{x}}_k^- &= f_{k-1}(\hat{\mathbf{x}}_{k-1}^+, \mathbf{u}_{k-1}^*) \\ P_k^- &= A_{k-1} P_{k-1}^+ A_{k-1}^\top + Q_{k-1} \end{aligned}$$

At the end of this step, compute the Jacobian matrix of the output equation

$$C_k = \left. \frac{\partial g_k(\mathbf{x}_k)}{\partial \mathbf{x}_k} \right|_{\mathbf{x}_k=\hat{\mathbf{x}}_k^-}$$

- **Step 2 (immediately after the time step k):** At the time step k , the measurement $\mathbf{y}_k$ arrives . Immediately after the time step k , we compute the Kalman gain matrix

$$K_k = P_k^- C_k^\top (C_k P_k^- C_k^\top + R_k)$$

Then, compute the a posteriori state estimate:

$$\begin{aligned}\hat{\mathbf{x}}_k^+ &= \hat{\mathbf{x}}_k^- + K_k(\mathbf{y}_k - C_k \hat{\mathbf{x}}_k^- - \mathbf{b}_k) \\ \hat{\mathbf{x}}_k^+ &= \hat{\mathbf{x}}_k^- + K_k(\mathbf{y}_k - C_k \hat{\mathbf{x}}_k^- - g_k(\hat{\mathbf{x}}_k^-) + C_k \hat{\mathbf{x}}_k^-) \\ \hat{\mathbf{x}}_k^+ &= \hat{\mathbf{x}}_k^- + K_k(\mathbf{y}_k - g_k(\hat{\mathbf{x}}_k^-)),\end{aligned}$$

where the third form of the above-stated equation is the most appropriate for computations. Finally, compute the a posteriori covariance matrix by using either this equation:

$$P_k^+ = (\mathbb{1} - K_k C_k) P_k^-$$

or

$$P_k^+ = (\mathbb{1} - K_k C_k) P_k^- (\mathbb{1} - K_k C_k)^\top + K_k R_k K_k^\top.$$

Due to the numerical stability, the last equation is a more preferable option to be used for covariance matrix time propagation.